

Systems Infrastructure and Security

Assessment Brief

Module Leader:	Fiona Redmond
Assessment Title:	System Security Project
Assessment Type:	Practical
Individual/Group:	Individual
Assessment Weighting:	50%
Hand-out Date:	9 th March 2026
Hand-in Date:	20 th April 2026 09:00am
Mode of Submission:	Documentation to be submitted on Blackboard. Demonstration on your scheduled day. NOTE: Virtual Machines not to be accessed past the deadline.

Project Overview

This project requires you to create an automated protection and monitoring environment using **Prometheus** (monitoring and alerting tool), **Grafana** (visual dashboard tool), and **Fail2ban** (IDS/IPS) on your Rocky Linux and Ubuntu servers. You must configure and demonstrate these tools effectively, ensuring proper functionality, alerting, visualisation, and documentation.

1. Environment Setup & Functionality (50 Marks)

Successful installation, configuration, and integration of:

Prometheus (15 marks):

- Collect data from 2 data sources.
- Configure 1 alert to notify administrators about security-related events.

Grafana (15 marks):

- Connect Grafana to Prometheus.
- Create 2 dashboards to visualise security metrics.

Fail2ban (15 marks):

- Monitor login attempts (e.g., SSH).
- Automatically block suspicious IP addresses based on predefined rules.

Overall System Stability & Effectiveness (5 marks)

System maintains security and responsiveness during testing.

2. Monitoring & Visualisation (20 Marks)

Dashboard Quality & Insight (10 marks)

- Effective and clear visualisation of key security metrics.
- Relevant and well-structured dashboards.

Alerting & Response (10 marks)

- Meaningful alerts based on security threats.
- Correct response and action implementation.

3. Documentation (15 Marks)

Setup and Configuration Steps (5 marks)

- Clear and reproducible steps. Include references where necessary.
- Screenshots and explanations where applicable.

Configuration Files & Customisation (5 marks)

- Inclusion of modified configuration files (e.g., prometheus.yml).

Security Analysis & Observations (5 marks)

- Critical reflections on system security.
- Identified improvements and limitations.

4. Demonstration (15 Marks)

Clarity and Effectiveness (10 marks)

- Concise, well-structured presentation within 9 minutes.
- Demonstration of security features in action (fail2ban, Prometheus alerts, Grafana dashboards).

Engagement and Understanding (5 marks)

- Confident and knowledgeable responses to questions

Focus on **key features** that demonstrate your understanding of the project and its impact on security. Demonstrations will take place during our labs in the final week. Schedule of demos can be found at the end of this brief. You must complete a demo for your submission to be accepted. i.e., your project will not be graded based on your virtual machines and documentation.

Submission

You must submit:

- A written report (via Blackboard).
- A functional system (demonstrated during your scheduled time).
- NOTE: Failure to demonstrate will result in no grade for the project and late submissions of report will incur a 5% penalty per day, up to a maximum of 7 days. Report submissions received more than 7 days after the deadline will not be accepted unless prior extensions are agreed upon.

Additional Details

Prometheus Configuration:

- Scrape metrics from at 2 data sources using exporters. Examples (but not limited to):
 - **Node exporter** for CPU, memory, and disk usage.
 - **Fail2ban exporter** for login attempts and blocked IPs.
 - Optional bonus:
 - Blackbox exporter for probing of endpoints over HTTP, HTTPS, DNS, TCP and ICMP.
 - MySQL Server exporter for monitoring and exposing MySQL metrics
 - Other exporters exist for services such as web servers and email. Think about what you have installed in labs such as ClamAV, Nikto etc.
- Set up **1 alert** using Prometheus Alertmanager to notify administrators of potential security events based on specific metric thresholds. Examples:
 - High number of failed login attempts.
 - CPU usage > 80%
 - Changes to critical files (e.g., /etc/passwd).
 - Consider how notifications will be sent (e.g. by email, Teams, slack channel, discord channel).

Grafana Configuration:

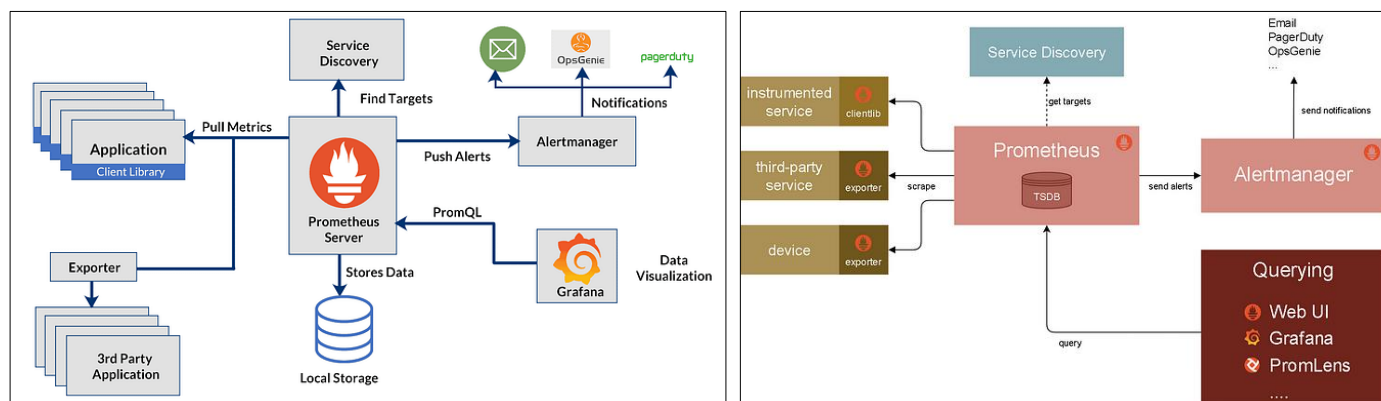
- Connect Grafana to the Prometheus server as a data source and provide real-time insights into system performance
- Create 2 dashboards to visualise metrics.
 - Dashboard 1
 - CPU
 - Memory
 - Disk
 - Failed login attempts
 - Dashboard 2
 - Any other security metric (e.g. Blocked IPs)
- You may use Grafana dashboard templates (e.g. template ID 14513) provided you understand the metrics shown, or you may build your own.

Fail2ban Integration:

- Configure Fail2ban to monitor (detection) a service like SSH.
- Set ban rules (prevention) based on criteria such as repeated failed logins.

Resources:

- <https://www.fosstechnix.com/install-prometheus-and-grafana-on-ubuntu-24-04/>
- <https://prometheus.io/docs/instrumenting/exporters/>
- <https://grafana.com/docs/versions/?project=%2Fdocs%2Fgrafana%2F&pg=webinar-getting-started-with-grafana-dashboard-design-amer&plcmt=related-content-2>



192.168.56.30:9090/targets?search=#pool-Node					
Prometheus Alerts Graph Status Help					
Targets					
All scrape pools All Unhealthy Collapse All Filter by endpoint or labels Unknown Unhealthy Healthy					
Node (1/1 up) show less					
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://192.168.56.30:9100/metrics	UP	instance="192.168.56.30:9100" job="Node"	10.280s ago	10.556ms	
prometheus (1/1 up) show less					
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	14.324s ago	6.145ms	

Assessment Criteria:

Breakdown	Mark
Environment Setup & Functionality	50
Monitoring and Visualisation	20
Documentation	15
Demonstration	15
TOTAL	100